

Smoking Cessation After Initial Treatment Failure With Varenicline or Nicotine Replacement

A Randomized Clinical Trial

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IMPORTANCE Most people who smoke do not quit on their initial attempt.

OBJECTIVE To determine the best subsequent strategy for nonabstinence following initial treatment with varenicline or combined nicotine replacement therapy (CNRT).

DESIGN, SETTING, AND PARTICIPANTS Using a double-blind, placebo-controlled, sequential multiple assignment randomized trial, 490 volunteers were randomized to receive 6 weeks of varenicline or CNRT. After 6 weeks, nonabstainers were rerandomized to continue, switch, or increase medication dosage for 6 additional weeks. The study was conducted from June 2015 through October 2019 in a Texas tobacco treatment clinic.

INTERVENTIONS The initial treatment was 2 mg/d of varenicline or the combined replacement therapy of a 21-mg patch plus 2-mg lozenge. The rerandomized participants either continued with their initial therapies, switched between varenicline and CNRT, or increased dosages either to 3-mg or more of varenicline or to a 42-mg patch and lozenges. All received weekly brief counseling.

MAIN OUTCOMES AND MEASURES Biochemically verified 7-day point prevalence abstinence at the end of treatment at 12 weeks.

RESULTS The 490 randomized participants (210 female [43%], 287 non-Hispanic White [58%], mean age, 48.1 years) smoked an average of 20 cigarettes per day. After the first phase, 54 participants in the CNRT group were abstinent and continued their therapy; of the 191 who were not abstinent, 151 were rerandomized, and the 40 who did not return for rerandomization were assigned to continue their initial CNRT condition in phase 2. The end-of-treatment abstinence rate for the 191 phase 1 nonabstainers was 8% (95% credible interval [CrI], 6% to 10%) for the 90 (47%) who continued at the dosage condition, 14% (CrI, 10% to 18%) for the 50 (33%) who increased their dosage, and 14% (95% CrI, 10% to 18%) for the 51 (34%) who switched to varenicline (absolute risk difference [RD], 6%; 95% CrI, 6% to 11%) with more than 99% posterior probability that either strategy conferred benefit over continuing the initial dosage. After the first phase, 88 participants in the varenicline group were abstinent and continued their therapy; of the 157 who were not abstinent, 122 were rerandomized and 35 who did not return for rerandomization were assigned to continue with the varenicline condition. The end-of-treatment abstinence rate for the 157 phase 1 nonabstainers was 20% (95% CrI, 16% to 26%) for the 39 (32%) who increased their varenicline dosage, 0 (95% CrI, 0 to 0) for the 41 (34%) who switched CNRT, and 3% (95% CrI, 1% to 4%) for the 77 (49%) who were assigned to the continued varenicline condition (absolute RD, -3%; 95% CrI, -4% to -1%) with more than 99% posterior probability that continuing varenicline at the initial dosage was worse than switching to a higher dosage. Furthermore, increasing the varenicline dosage had an absolute RD of 18% (95% CrI, 13% to 24%) and a more than 99% posterior probability of conferring benefit. The secondary outcome of continuous abstinence at 6 months indicated that only increased dosages of the CNRT and varenicline provided benefit over continuation of the initial treatment dosages.

CONCLUSIONS AND RELEVANCE For individuals who smoked but did not achieve abstinence after treatment with varenicline, increasing the dosage enhanced abstinence vs continuing, whereas for nonabstainers initially treated with CNRT, a dosage increase or switch to varenicline enhanced abstinence and may be viable rescue strategies.

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Quitting smoking reduces the prevalence, morbidity, and/or mortality of tobacco-related cancers, stroke, cardiovascular disease, diabetes, chronic obstructive pulmonary disease, and rheumatoid arthritis and reduces their associated health care expenditures.^{1,2} National surveys suggest that annually, nearly 70% of people who smoke want to quit smoking,³ and 55% report having made a serious attempt to quit (no smoking for at least 24 hours). However, only about 7.5% succeed⁴ and only 4.7% receive recommended,⁵ evidence-based counseling and pharmacotherapy.³ Most require multiple quit attempts (≈ 6) to achieve long-term success,⁶ but there is little evidence to guide subsequent treatment strategies. The current trial used a sequential, multiple assignment, randomized trial (SMART)^{7,8} design to rerandomize nonabstainers after an initial treatment failure to alternative pharmacological approaches to address this need.

Several approved smoking cessation medications exist, including varenicline, bupropion, and nicotine replacement therapy products.^{5,9-13} Few studies have examined treatment options after an initial treatment failure. Among those studies, the initial treatment has been limited to a single nicotine patch,^{14,15} which is less effective than combined nicotine replacement (nicotine-patch + lozenge or gum),^{16,17} thereby attenuating the initial treatment response. On average, varenicline and combined nicotine replacement therapy (CNRT) appear to consistently produce the highest levels of long-term abstinence than other pharmacotherapies for cessation.^{17,18} No studies have determined the optimal strategy for patients who have not quit or relapsed after their initial treatment with these medications, although one observational study reported increased abstinence by increasing the dose to 3 mg among those who had reduced but had not quit,¹⁹ whereas another larger trial showed no such benefit.²⁰

Taking the perspective of a clinician in the current environment in which both varenicline and CNRT are recommended first-line treatments,^{21,22} this study addressed the following questions: First, what is the probability of abstinence between these 2 treatments on an initial quit attempt? Second, what is the best subsequent treatment approach for those nonabstinent after initial treatment? Relative to continuing the same medication strategy, the hypothesis of this study was that nonabstainers would derive the most benefit by increasing the varenicline dosage after initially taking varenicline or switching to varenicline from a CNRT strategy, anticipating similar abstinence rates between these 2 strategies at the end of 6 weeks.

Methods

Trial Design

Figure 1 depicts the SMART design. In phase 1 (weeks 1-6), participants were randomized to receive either the standard dosage of varenicline or CNRT. In phase 2 (weeks 7-12), phase I abstainers (7-day point-prevalence with expired carbon monoxide [CO] <6 ppm) continued their medication. Nonabstainers were rerandomized to (1) continue their phase 1 medication dosage, (2) switch to the alternate phase 1 medication, or (3) increase the phase 1 medication dosage for another 6 weeks.

Key Points

Question What is the best treatment strategy after initial treatment failure for smoking cessation?

Findings Participants receiving varenicline or combination nicotine replacement therapy who were nonabstinent after 6-weeks were rerandomized to continue their medication, increase the dosage, or switch medications, for another 6-weeks. Increasing the dosage led to the greatest benefit for those originally taking varenicline, whereas those receiving combined nicotine replacement therapy benefited similarly from the dosage increase and switching to varenicline, although secondary abstinence measures suggested that the dosage increase may provide longer-term benefit.

Meaning This study provides clinicians with guidance on the best rescue strategies for nonabstinence following an initial quit attempt.

Rerandomizing nonabstainers to continue the same medication represented the typical clinical approach of simply allowing more time on the same medication. All treatments included the corresponding placebo for the alternate medication (study protocol [Supplement 1](#)).

Participants

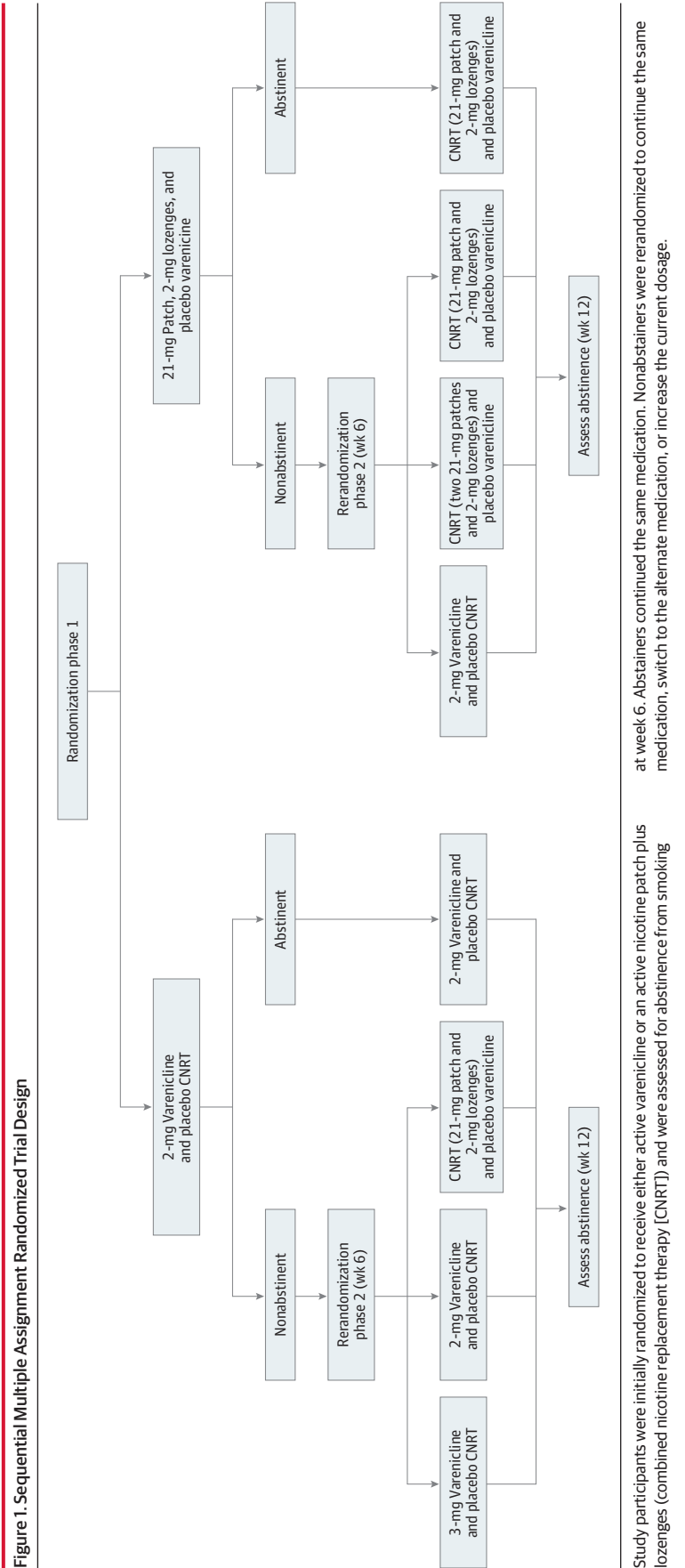
Participants were recruited from the Houston, Texas, area using media advertising from June 2015 through October 2019. Participants were included if they were aged 18 to 75 years; smoked 5 or more cigarettes per day with an expired CO of 6 ppm or more; were fluent in English; had stable housing and a phone; and provided written consent. People were excluded if they had used other smoking treatments or medications contraindicated for varenicline or CNRT; had a moderate or higher suicidality score on the Mini International Neuropsychiatric Interview [MINI]²³; met criteria for current or lifetime mania; were psychotic; had a substance use disorder; or had other unstable psychiatric disorders. The University of Texas MD Anderson Cancer Center Internal Review Board approved this research.

Interventions

Pharmacotherapy

Phase 1 pharmacotherapy was initiated the day after randomization, 7 days prior to the quit date. In the varenicline group, dosing followed the recommended titration schedule (ie, 0.5 mg/d varenicline for days 1-3; 0.5 mg twice a day for days 4-7; and 1 mg twice a day thereafter). In the CNRT condition, dosing consisted of a 21-mg patch plus 2-mg lozenges (taken as needed) with a recommended use of at least 6 lozenges per day.

In phase 2, participants with carbon monoxide-verified 7-day point prevalence abstinence at week 6 measured at an in-clinic visit continued their phase 1 medication through the end of treatment at week 12. Phase 1 nonabstainers were then randomized to 1 of 3 medication conditions: (1) continue their phase 1 treatment; (2) switch medication conditions that followed the phase 1 dosing levels, and (3) increased dosages for either medication condition. The increased varenicline dosage was implemented by taking an extra 1-mg tablet (3-mg total daily dosage) with the evening dose, and the CNRT was accomplished by applying an additional 21-mg nicotine patch



(42 mg total daily), in the morning as usual, plus continuing the 2-mg lozenges taken as in phase 1. Inactive placebos used the same regimen as corresponding active medications. Dose adjustments by the blinded study physician were permitted to control adverse events.

Behavioral Counseling

All participants received brief counseling (15 minutes) conducted over 2 in-clinic and 3 telephone visits in phase 1, and 2 in-clinic and 1 telephone visit in phase 2, similar to previous trials²⁴⁻²⁶ (see eAppendix 1 in [Supplement 2](#)).

Follow-Up

Follow-ups were conducted 30 days after the end of treatment and 6 months after the initial quit day.

Outcomes

Self-report of smoking was assessed using the timeline follow-back procedure.^{27,28} The prespecified primary outcome for this trial was self-reported 7-day point prevalence abstinence (ie, self-report of no smoking in previous 7 days) plus expired CO of less than 6 ppm at end of treatment (week 12). The secondary outcome of continuous abstinence from the end of treatment was assessed at follow-up. Adverse events were assessed using the Common Terminology Criteria for Adverse Events version 4.03.²⁹

Randomization

Adaptive randomization (minimization)³⁰ was used for the phase 1 group assignment to stratify the sample by gender and race using a computer program generated by the study data team.

Blinding

All staff were blinded, except the study coordinator who prepared the kits for distribution. Active nicotine replacement therapies and matching placebos were purchased from NAL Pharma. The clinical image of varenicline and placebo were provided by Pfizer. The University of Pennsylvania Investigational Drug Service prepackaged daily dosages of all medications into coded-blinded blister packs and bottles (lozenges).

Sample Size Determination and Statistical Analysis Plan

Details for sample size determination and point estimates are provided in eAppendix 2 and eFigure 1 in [Supplement 2](#). We used a bayesian approach³¹ to provide unbiased estimates for the benefit conferred by the switch and increased dosage conditions, relative to staying within each primary treatment group. For each comparison, we estimated the probability of detecting a benefit at 0.80 or greater using $K = 1000$ Monte Carlo simulations (see eTables 1 and 2 in [Supplement 2](#)). All comparisons demonstrated an 80% or more chance of detecting a more than 80% probability of benefit for a total of 500 participants, which we prespecified as the posterior probability sufficient for confirmation of our hypotheses. Given our stated a priori goal of providing precise, unbiased probability estimates for each treatment, primary outcomes analyses did not adjust for multiplicity.

Bayesian methods have been discussed previously by the US Food and Drug Administration as an alternative to the frequentist approach for making clinical decisions and evaluating new treatments.³² Our approach was to model the sequential decision-making of clinicians, ie, deciding the initial treatment, observing the response, and taking a subsequent course of action. By design, this approach did not focus on determining the single best treatment pathway beginning with the initial treatment. We reasoned that in today's clinical environment both CNRT and varenicline remain viable first-line treatments, for which the choice may be based on cost, availability, previous experience, patient-clinician preference, in addition to efficacy. Hence, comparisons within initial treatment arms best fit the current clinical milieu.

Logistic regression was used to evaluate abstinence among treatments. Prior specification for logistic models took the form: normal distribution ($\sim N[\mu = 0, \sigma = 1000]$) in the log (odds) scale. Inverse probability weighting (for the probability of assignment to a specific sequence of treatments by design) permitted unbiased effect size estimates in the context of rerandomization. All analyses used R version 4.0.3.³³

Results

Baseline Characteristics and Participant Flow

The [Table](#) and eTable 3 in [Supplement 2](#) provide sample characteristics by phase 1 and phase 2 conditions, respectively. The 490 participants (210 women [43%], 287 non-Hispanic White (58%), mean age 48.1 years, and >79% employed) averaged 20 cigarettes per day. [Figure 2](#) shows the disposition of participants through the trial. A total of 491 participants were randomized in phase 1 to CNRT and varenicline. One person in the CNRT group was eliminated after randomization (leaving 490) due to a protocol violation, rendering their data unusable. At the conclusion of phase 1 (week 6), there were 191 nonabstainers in the CNRT group and 157 in the varenicline group who were rerandomized to continue their respective conditions, switch, or increase their medication dosages. Among those, the 40 participants in the phase 1 CNRT group and the 35 in the varenicline group who did attend rerandomization were assigned to continue treatment as initially randomized for the phase 2 analyses. This is consistent with approaches used in other clinical trials for smoking cessation of including anyone initially randomized to a treatment to be among those evaluated for that treatment once exposed. Of those who had abstained, 54 in the CNRT group and 88 in the varenicline group continued their phase 1 medications for phase 2. At the conclusion of phase 2 (end of treatment at week 12), all participants were analyzed for the primary outcome of 7-day point prevalence abstinence. [Figure 2](#) shows the number of participants lost to follow-up and analyzed as nonabstinent (intent to treat) across conditions.

Outcomes and Estimation

Smoking Cessation End of Phase 1

At week 6, the probability of abstinence for those randomized to CNRT was 22% (95% credible interval [CrI], 17%-27%;

Table. Baseline Measures and Demographics^a

Variable	No. (%) of participants	
	Combined nicotine replacement therapy (n = 245) ^b	Varenicline (n = 245)
Demographics		
Age, mean (SD), y	48.9 (11.4)	47.37 (11.2)
Sex		
Female	105 (42.9)	105 (42.9)
Male	145 (57.1)	145 (57.1)
Race and ethnicity, No. (%) ^c		
African American, non-Hispanic	64 (26.1)	65 (26.5)
Asian, non-Hispanic	9 (3.7)	3 (1.2)
Hispanic, any race	19 (7.8)	22 (9)
>1 Race	5 (2.0)	6 (2.4)
White, non-Hispanic	143 (58.3)	144 (58.8)
Other	5 (2.0)	5 (2.0)
Employment		
Yes	181 (73.9)	179 (73.1)
No	64 (26.1)	66 (26.9)
Education, median (IQR), y	13 (12-14)	13 (12-14)
Household income, \$		
<50 000	115 (46.9)	107 (43.7)
50 000-79 999	51 (20.8)	68 (27.8)
≥80 000	73 (29.8)	63 (25.7)
Not reported	6 (2.4)	7 (2.9)
Smoking characteristics		
CO, median (IQR), ppm ^d	19 (13-26)	21 (15-28)
Cigarettes per d, median (IQR)	20 (12-20)	20 (11.8-20)
FTCD, median (IQR), total score ^e	5 (3-6)	5 (3.8-6)
Years of smoking, mean (SD)	28.6 (12.5)	28.0 (11.8)
Age of smoking initiation, median (IQR)	17 (14-20)	17 (14-19)

Abbreviations: CO, carbon monoxide; FTCD, Fagerström test for cigarette dependence.

^a Additional demographic information by phase 2 randomization groups can be found in eTable 3 in Supplement 2.

^b The combined nicotine replacement therapy was 21-mg nicotine patch plus 2-mg lozenges.

^c National Institutes of Health categories for the self-reported race and ethnicity were used.

^d A CO level of 6 ppm or higher was required for eligibility.

^e A 6-item instrument with a possible range of 0 to 10 with higher scores meaning more cigarette dependence. In general, a score of 5 or higher is considered moderate to severe dependence.

54 of 245) and was 36% (95% CrI, 30%-42%; 88 of 245) for those randomized to varenicline. The absolute risk difference (RD) of -14% (95% CrI, -22% to -6%) between these conditions was associated with a posterior probability of 99% or more of a non-zero difference. This result was contrary to our initial assumption of similarity between CNRT and varenicline abstinence rates at 6 weeks.

Smoking Cessation End of Phase 2

Among week-6 nonabstainers, we evaluated the impact of each rerandomization strategy on abstinence at end of treatment (week 12) relative to continuing the same medication. As shown in Figure 3, among 191 in the CNRT who had not abstained by week 6, end of treatment abstinence probabilities resulted in 14% (95% CrI, 10%-18 %; 7 of 51) among those who were switched to varenicline; 14% (95% CrI, 10%-18%; 7 of 50) among those whose CNRT had been increased, and 8% (95% CrI, 6%-10%; 7 of 90) among those who continued at the initial CNRT dosages. Relative to continuing with the initial CNRT dosage, switching to varenicline or to increased dosages of the CNRT, both resulted in an absolute RD of 6% (95% CrI, 2%-11%) and a posterior probability of more than 99% that either strategy conferred a nonzero benefit over CNRT continuation.

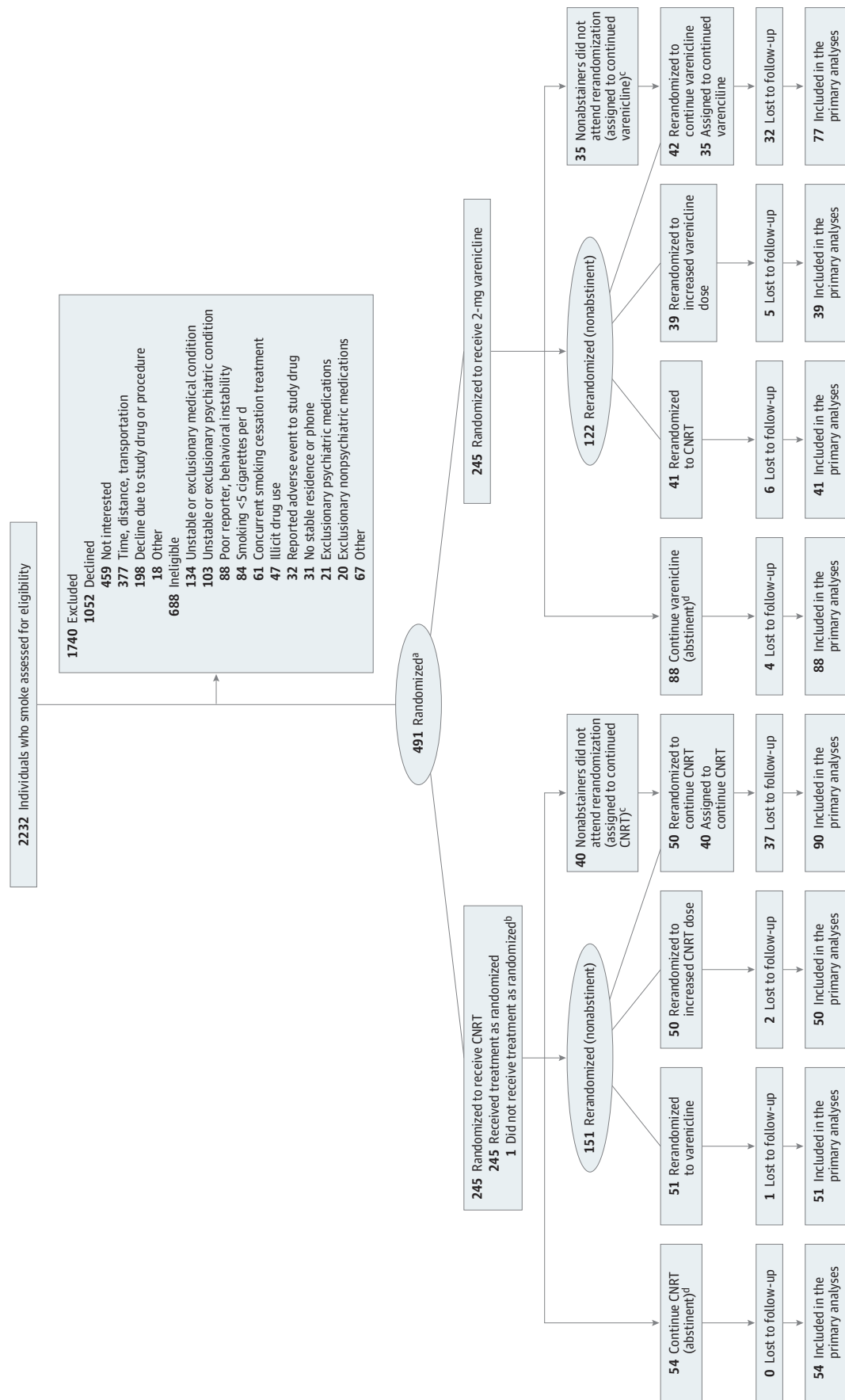
As shown in Figure 3, among the 157 participants randomized to varenicline who were nonabstinent at week 6, end of treatment abstinence probabilities at week 12 was 20% (95% CrI, 16%-26%; 8 of 39) among those who had increased their varenicline dosage; 0% (95% CrI, 0%-0%; 0 of 41) among those who were switched to CNRT; and 3% (95% CrI, 1%-4%; 2 of 77) among those who continued their initial varenicline dosage.

Relative to continuation on varenicline, switching to CNRT resulted in an absolute RD of -3% (95% CrI, -4% to -1%) and associated posterior probability of more than 99% that continuing was worse than switching. In contrast, increasing the varenicline dosage resulted in an absolute RD of 18% (95% CrI, 13%-24%) and posterior probability of more than 99% that increasing the dosage was superior to continuing. The number needed to treat for each contrast is presented in eTable 4 in Supplement 2.

Smoking Cessation End of Phase 2 Among Phase 1 Abstainers

At the end of phase 1, 54 abstainers in the CNRT group and 88 abstainers in the varenicline group continued their initial medication to week 12. As shown in Figure 3, the posterior probability of abstinence at week 12 among CNRT week 6 abstainers was somewhat higher at 78% (95% CrI, 69%-85%; 42 of 54)

Figure 2. The Flow Diagram Depicting Participant Flow Through the Trial



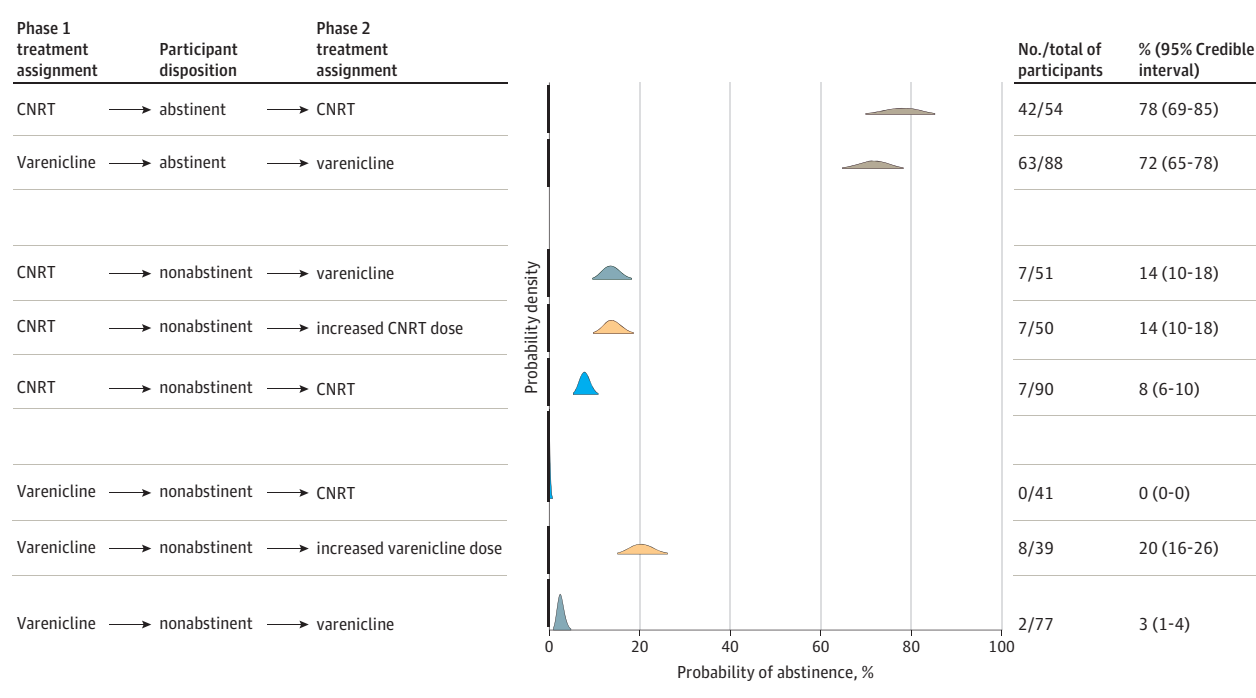
^aRandomization in phase 1 used minimization to balance gender and race.

^bParticipant was assigned to the wrong phase 2 condition and was excluded from all analyses.

^cConsistent with an intent-to-treat approach, participants were imputed into the continuation condition in phase 2.

^dAbstinence was 7-day point prevalence biochemically verified; continuous abstinence was measured 30 days after treatment and 6 months after the target quit date.

Figure 3. Posterior Probability Distributions for Abstinence at End of Treatment (12 Weeks)



Each distribution is a probability density with an area under the curve equal to 1. The height of the density is scaled to permit full depiction of each distribution. The phase 1 dose for the combined nicotine replacement therapy (CNRT) was a 21-mg patch and 2-mg lozenges and the varenicline dose was 2 mg/d. The phase 2 CNRT dose increased to 2 patches and lozenges remained at 2 mg, and

the varenicline dose increased to 3 mg/d. The color brown indicates participants who had abstained after 6 weeks; lighter blue, nonabstainers who switched treatment; orange, nonabstainers whose dosages were increased; and darker blue, nonabstainers who continued phase 1 treatment.

than week 6 varenicline abstainers at 72% (95% CrI, 65%-78%; 63 of 88); the posterior probability of a difference between these 2 conditions was 88% (absolute RD, 6%; 95% CrI, -5% to 16%). This difference was unexpected.

Adverse Events

Estimated frequencies and 95% CrIs, for adverse events for phases 1 and 2 are presented in eTables 5 and 6 in [Supplement 2](#), respectively. There were no adverse events for which group differences exceeded 2% with nonoverlapping CrIs, that were attributable to medication, except for nausea, which was higher in the phase 1 varenicline group vs the CNRT group. Three serious adverse events were recorded as unrelated to treatment.

Visit and Medication Adherence

Visit and medication adherence are shown in eTables 7 and 8 in [Supplement 2](#) for phases 1 and 2, respectively. Visit attendance was generally higher in phase 1 than phase 2 and higher for abstainers than nonabstainers in phase 2. No major differences were noted for pill or patch adherence: the mean prescribed dosage taken averaged 82% or higher for active varenicline and 77% or higher for the active CNRT patch. Combined nicotine replacement therapy active lozenge count was more variable, ranging from a median of 40 lozenges (IQR, 0-118) to 76 lozenges (IQR, 9.75-140) across phases.

Secondary Outcomes of Abstinence

Secondary outcomes of continuous abstinence at the end of treatment plus 30-day and the 6-month posttarget quit-date follow-up points are described in eAppendix 4 in [Supplement 2](#). These are considered descriptive and collaborative of our primary outcomes, without correction for multiplicity. Posterior probability distributions for continuous abstinence measured at 30 days after treatment are shown in eFigure 2 in [Supplement 2](#). Among 191 nonabstainers in phase 1 in the CNRT group, both the dosage increase (8%; n = 50) or the switch to varenicline (10%; n = 51) provided more benefit (posterior probability >99%; see eTable 9 in [Supplement 2](#) for absolute RD and 95% CrI) than continuing the initial treatment (3%; n = 90). However, for phase 1 nonabstainers in the varenicline group (n = 157), only increasing its dosage (8%; n = 39) provided benefit relative to continuation (0%; n = 42). When measuring 6-month continued abstinence (eFigure 3 in [Supplement 2](#)) among phase 1 nonabstainers in both groups, only the CNRT patch increase (3%) and varenicline dosage increase (2%) provided benefit over continuation (0%) of the respective therapies with probabilities of 96% and 99%, respectively of a non-zero difference (see eTable 9 in [Supplement 2](#)).

eTable 10 in [Supplement 2](#) shows the absolute RDs and probabilities for the phase 2 increase vs respective switch conditions contrasts for both CNRT and varenicline phase 1 nonabstainers at each time point.

When measured 30 days after the end of treatment, a higher proportion of continuous abstinence was observed among phase 1 abstainers in the CNRT group than in the varenicline group (probability >97%), but no differences were observed at 6 months (see eFigures 2 and 3 and eTable 11 in Supplement 2).

Post Hoc Analyses

For our primary outcome only, we also estimated the average probability of week-12 abstinence for each treatment pathway by combining those who had abstained in phase 1 and continued their treatment with those who had not abstained and were rerandomized to 1 of the 3 alternatives. The results are consistent with those reported above when comparing individual conditions. The highest abstinence rate among those in the varenicline group was achieved when combining week-6 abstainers with nonabstainers who received a dosage increase. Among those starting in the CNRT group, similar abstinence rates were observed when week-6 abstainers who continued their therapy were combined with nonabstainers whose dosages were increased or who had been switched to varenicline (eAppendix 5 and eTable 12 in Supplement 2).

Discussion

Among week-6 CNRT nonabstainers similar increases were observed in 7-day posterior probability of abstinence of 6% by week 12 between those rerandomized to continue CNRT (8%) vs either switching to varenicline (14%) or increasing the dosage of CNRT (14%). However, among the nonabstainers in the varenicline group at 6 weeks, a 17% abstinence increase was observed at week-12 among those rerandomized to varenicline (3%) vs those rerandomized to the increased dosage of varenicline (20%), but a 3% decrement of 3% was observed between those rerandomized to varenicline (3%) and switching to CNRT (0%). Relative to continuing the same medication, no evidence was found for increased risk of adverse events with the dosage increase for either medication.

Secondary outcomes of continuous abstinence 30 days after the end of treatment and after 6 months (see Supplement 2) followed a similar pattern, although absolute differences in abstinence between conditions are lower. However, only the dosage increase conditions for either treatment strategy for those who did not abstain at week 6 provided the best benefit at 6 months. Thus, either switching to varenicline or increasing the dosage of nonabstainers initially treated with CNRT may provide an initial benefit, even though increasing the CNRT may be better in the longer-term, whereas increasing varenicline to 3 mg for those who did not abstain at the 2-mg dose may provide both initial and longer-term benefit relative to both continuation at the lower dosage or switching to CNRT. The results at the distal follow-ups should be interpreted with caution and require replication in a larger sample.

The probability of abstinence at the end of phase 1 was 22% for those in CNRT group and 36% for those in the varenicline group.

This was unexpected. Abstinence among those in the CNRT group was lower than that reported by Baker and colleagues³⁴ ($\approx 36\%$ for both varenicline and CNRT groups) at a similar time point (4 weeks). A randomized clinical trial (RCT),³⁵ a meta-analysis,³⁶ and a population-based study³⁷ also showed slightly higher abstinence rates among those taking varenicline compared with CNRT at similar times; however, a recent Cochrane review showed a small advantage for varenicline over a full course of medication.³⁸ All these trials were published after this trial's inception. A network analysis available during the planning of this study showed no differences between those 2 medication groups,¹⁶ which was our expectation.

Interestingly, the posterior probability of abstinence at week 12 among those who had abstained at week 6 and continued CNRT was somewhat higher than among those who had abstained at week 6 and continued taking varenicline. However, the absolute numbers of abstainers at the end of treatment favored varenicline ($n = 63$) over CNRT ($n = 42$) given the substantial difference in the week-6 quit rates: varenicline (36%); CNRT (22%). Secondary outcomes of continuous abstinence at 6 months, suggests that early quitters (week-6 abstainers) may benefit equally over the long-term from their initial medication.

Limitations

First, phase 1 nonabstainers who did not return for rerandomization were assigned to the continue condition of their respective medications because they were only exposed to 1 treatment strategy. It will be important in future studies to improve the rate of rerandomization particularly when keeping participants who continue taking their original medication as a control group for subsequent treatment approaches. Second, use of the 2-mg vs 4-mg lozenges may have limited the effectiveness of the CNRT approach for some people who smoke more heavily. This was a practical choice for this trial to limit the complexity of the dosing regimen given the patients' ability to self-titrate lozenge consumption and allow for better tolerability. Third, increasing the number of 21-mg patches to 2 or increasing varenicline to 3 mg, although generally noted as safe and previously used clinically,^{39,40} is considered off-label prescribing. Fourth, demonstrating the longer-term benefits of these treatment strategies in a future study may require a larger sample size. Follow-up trials with more rescue therapy alternatives are needed to provide broader treatment options in clinical practice.

Conclusions

For individuals who do not achieve abstinence after treatment with varenicline, increasing the dosage enhances abstinence relative to continuing at the initial dosage, whereas for those who don't achieve abstinence with CNRT, a dosage increase or switching to varenicline are viable rescue strategies, with continuous abstinence results favoring a dosage increase.

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Acquisition, analysis, or interpretation of data: Cinciripini, Green, Minnix, Robinson, Cui, Kim, Kypriotakis, Beneventi, Blalock, Versace, Karam-Hage.

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Supervision: Cinciripini, Minnix, Karam-Hage.

Other - medical aspects of the study: Karam-Hage.

Other - Support for participant screening and management: Blalock.

Other - Screened and managed participants with mental health disorders: Beneventi.

Conflict of Interest Disclosures: Dr Cinciripini reported that Pfizer provided medication for an National Institutes of Health (NIH) supported clinical trial outside the submitted work. Dr Minnix reported personal fees from UpToDate to author a chapter about treating alcohol and tobacco use in cancer patients outside the submitted work. Dr Robinson reported receiving grants from the NIH during the conduct of the study and outside the submitted work. Dr Cui reported receiving grants from the NIH during the conduct of the study. Dr Versace reported receiving grants from NIH during the conduct of the study. No other disclosures were reported.

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